AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. Complete the given table by stating the percentage of oxygen and carbon dioxide gas in the exhaled air of human beings. (2 Marks)

Gas	Inhaled Air (%)	Exhaled Air (%)
Oxygen	21	
Carbon dioxide	0.04	

b. The diagrams below show two states of the pupil of a human eye.



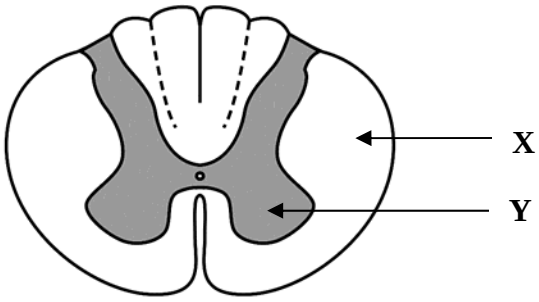
A **B**

Which diagram, **A** or **B**, shows the state of pupil of a person looking at a bright object? Give a reason to support your answer. (2 Marks)

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Q.2. (Total 4 Marks)

The given diagram shows cross-section of the spinal cord of human beings.



a. Name the areas, **X** and **Y** in the given diagram. (2 Marks)

b. Which parts of nerve tissues are present in **X** and **Y**? (2 Marks)

Q.3. (Total 4 Marks)

a. Name the type of connective tissue present in the human body that connects: (2 Marks)

i. muscles to bones.

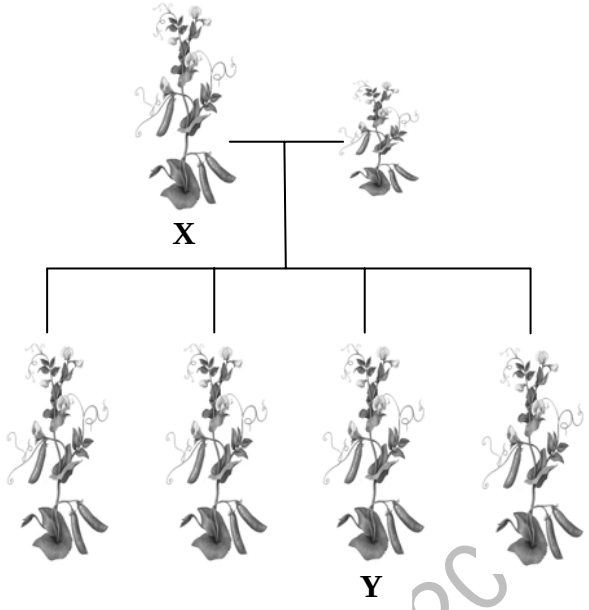
ii. bones to bones.

b. Mention TWO components of shoulder joint that make the arm of a human being move in all directions. (2 Marks)

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Q.4. (Total 4 Marks)

a. The given diagram shows a genetic cross between two pea plants of different heights. (2 Marks)



Determine the genotype of plant **X** and **Y**.

Genotype of **X**: _____

Genotype of **Y**: _____

b. (2 Marks)

Photosynthesis is the process in which plants, algae and some bacteria use the energy from sunlight to produce glucose from carbon dioxide gas and water.
In plants, carbon dioxide gas enters the cells through stomata. Water enters the roots from the soil and is transported up to the leaves through specialised plant tissues known as xylem vessels.

Identify any TWO biotic and TWO abiotic factors mentioned in the given information.

Biotic Factor	Abiotic Factor

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Q.5. (Total 4 Marks)

a. Which TWO events of meiosis lead to variation in living organisms? (2 Marks)

_____

_____

b. How do the events identified in part **a** lead to variation? (2 Marks)

_____

_____

_____

_____

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Q.6. (Total 5 Marks)

a. How has genetic engineering improved the quality and quantity of agricultural yield? (2 Marks)

i._____

_____

ii. _____

_____

b. Ali's classmate is suffering from influenza. As a preventive measure, Ali's mother got him vaccinated.

Describe the mode of action of vaccine that prevents Ali from getting influenza. (3 Marks)

_____

_____

_____

_____

_____

_____

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Q.7. (Total 7 Marks)

EITHER

a. Relate the onset of the following diseases of the lungs and the circulatory system with cigarette smoking.

i. Lung Cancer (2 Marks)

ii. Emphysema (3 Marks)

iii. Atherosclerosis (2 Marks)

OR

b. In a normal adult, about 120 cm^3 of filtrate is formed in the kidneys every minute. As this filtrate passes through the tubules, selective reabsorption occurs.

i. What will happen if this amount of filtrate were allowed to pass out as urine? (2 Marks)

ii. Describe the process of filtrate formation in nephrons of the human kidneys. (5 Marks)

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Q.8. (Total 8 Marks)

EITHER

a. Explain the process of vegetative propagation in onion and ginger. (8 Marks)

OR

b.

i. Describe any ONE example of predation. (2 Marks)

ii. Explain any TWO types of symbiotic interactions with an example for each type. (6 Marks)

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END OF PAPER

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